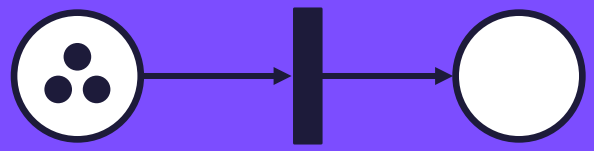
Chapter 2

Petri Nets: draw it, play it, break it



Modeling & Simulation of Networks · Master 1 RSD

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Warm-up: why not a flowchart?

⏱ 3 min



Two ambulances arrive. There is ONE free operating room. Who goes in?

Flowchart says…
"Step 1, step 2, step 3."
Great for one person following a recipe.
But it cannot show two patients competing for one room.

Petri net says…
Rooms are tokens. Patients are tokens. An event happens only if everything it needs is available.
Concurrency and sharing become visible.

What is a Petri net?

⏱ 3 min

A drawing of a system where things happen at the same time and share resources, invented by Carl Adam Petri (1962). You can draw it AND analyse it.

🏥 Healthcare

Patients, beds, doctors.
Can two patients take the same bed at once? No.

💻 IT

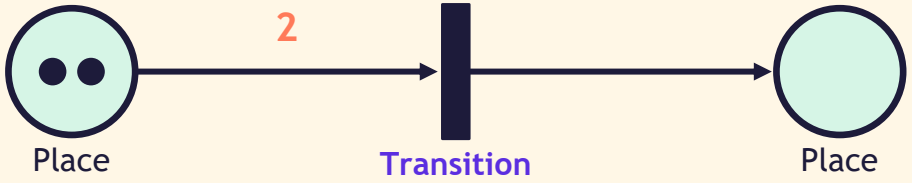
Processes and locks.
Can two programs write the same file at once? No.

🏭 Manufacturing

Machines, parts, repairmen.
What happens when a machine breaks? A repair loop.

Four shapes to remember

⏱ 4 min



Place (circle)
a state or a resource: "patients waiting", "free slots"

Transition (bar)
an event: "start", "send", "repair"

➜ Arc
links a circle to a bar and back; a number on it is the weight

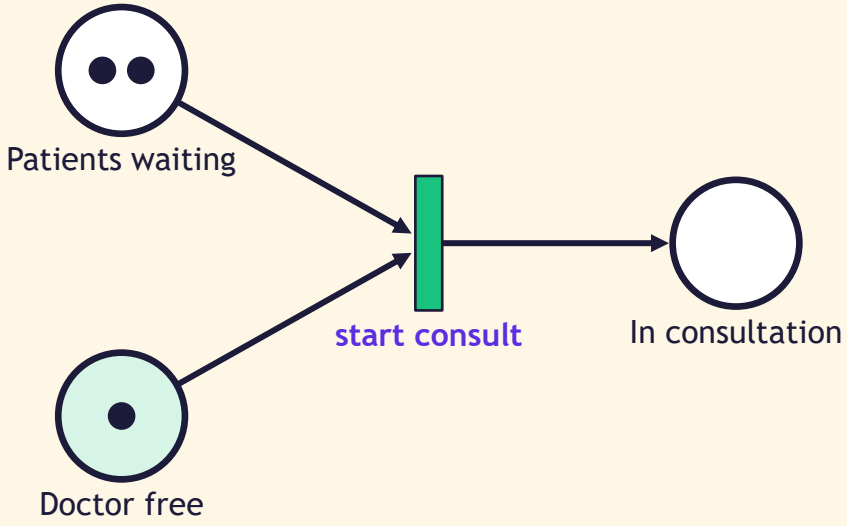
• Token
one unit that sits in a place. All tokens together = the marking

🤔 *Rule: circle to bar to circle. Never circle to circle!*

The firing rule: how tokens move

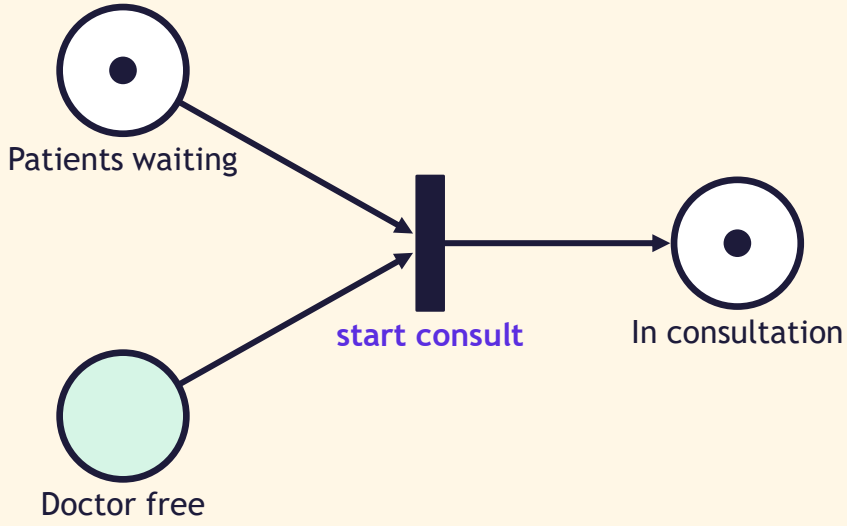
⏱ 4 min

BEFORE



FIRE!

AFTER FIRING

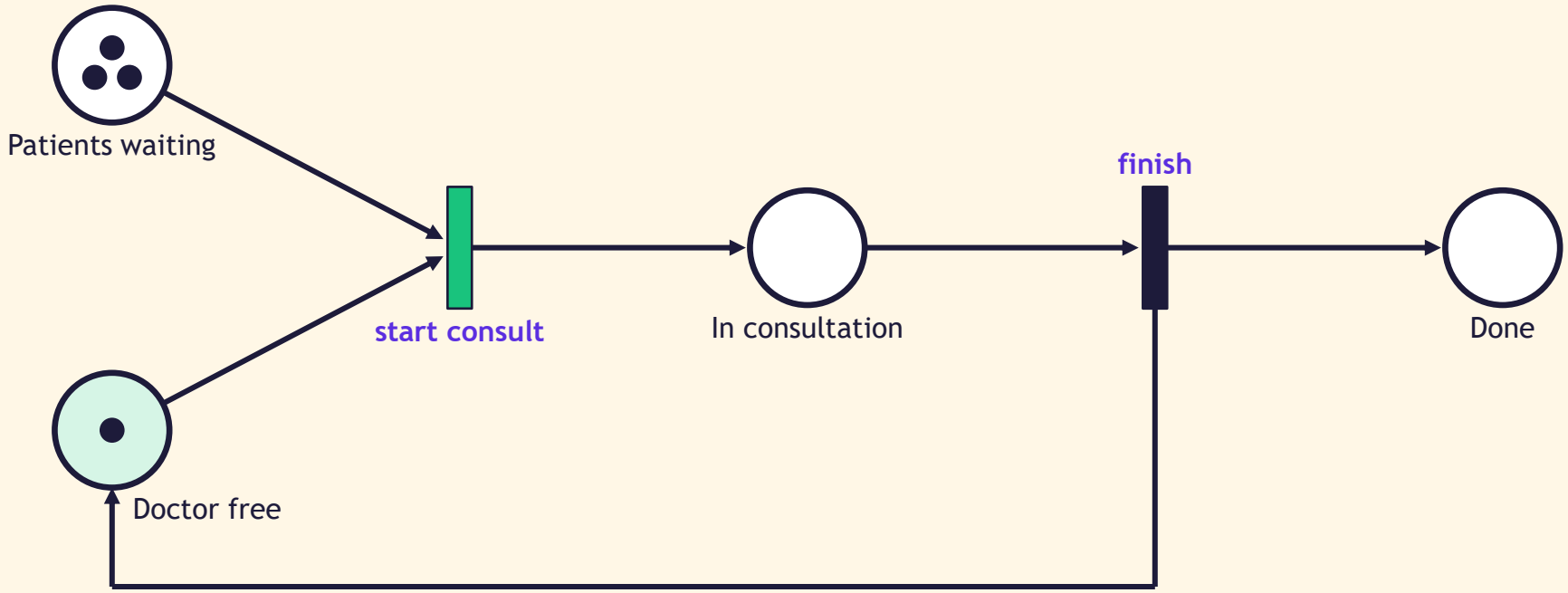


1 Enabled? 2 Take 3 Give

1) Does every input circle hold enough tokens? (green bar = yes) 2) Remove tokens from the inputs. 3) Add tokens to the outputs.

Example 1: the hospital clinic

⏱ 4 min



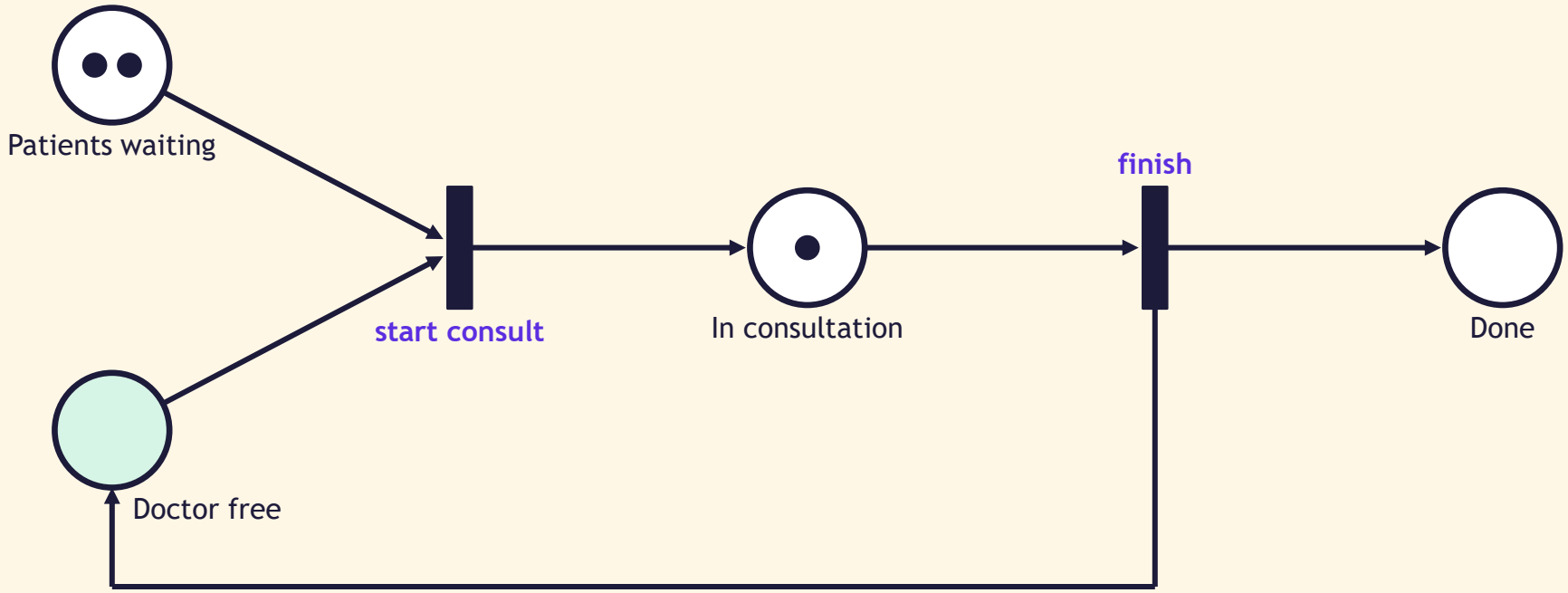


One doctor token = one doctor. The loop at the bottom gives her back after each patient.

Quiz 1: hospital — your turn

⏱ 3 min

Marking: 2 waiting, doctor busy, 1 consulting. Which transition is enabled (green)?



A. Start only

B. Finish only

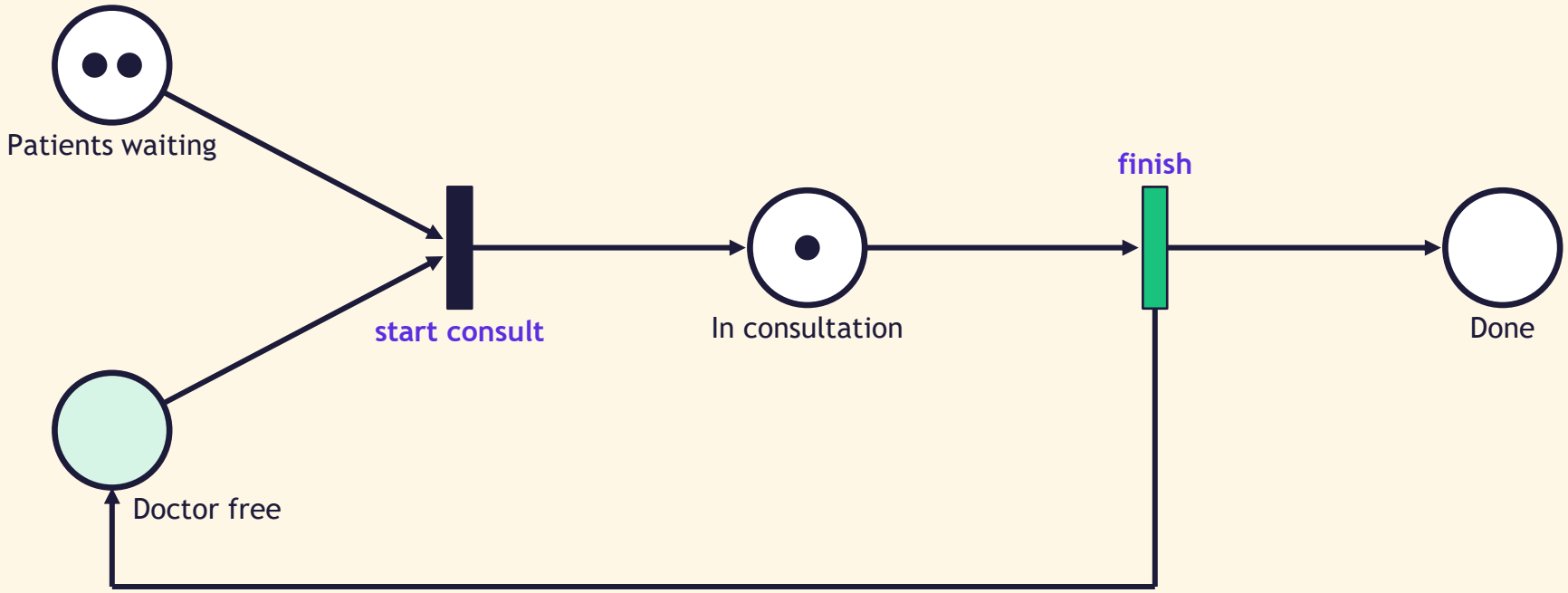
C. Both

D. None

Quiz 1: hospital — answer

⏱ 0 min

✅ Answer B: Finish only

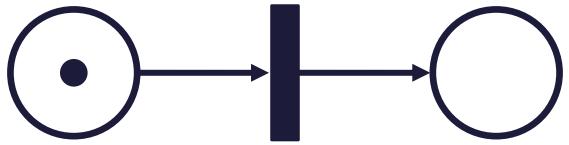


Start needs a free doctor: there is none. Finish needs a patient in consultation: yes. Only Finish can fire.

Four building blocks

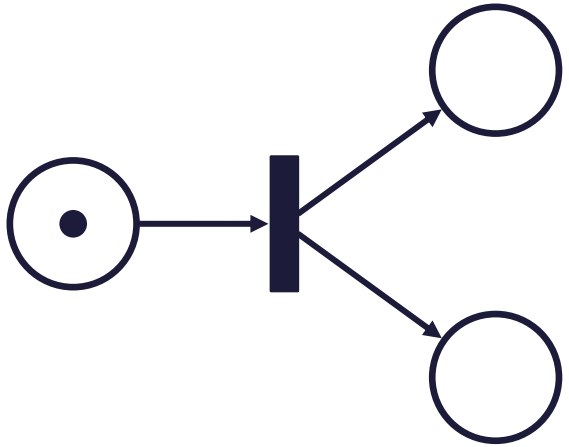
⏱ 5 min

Sequence



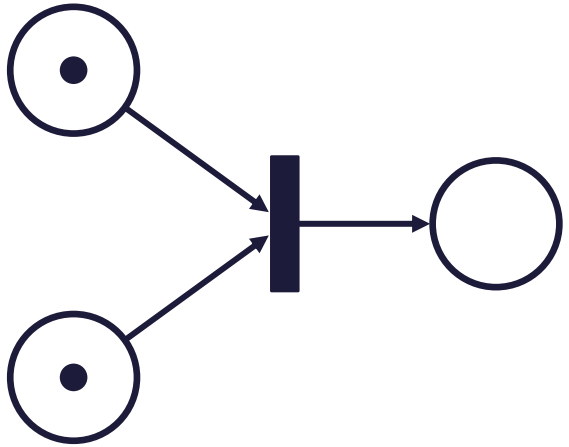
triage, then exam

Fork (AND)



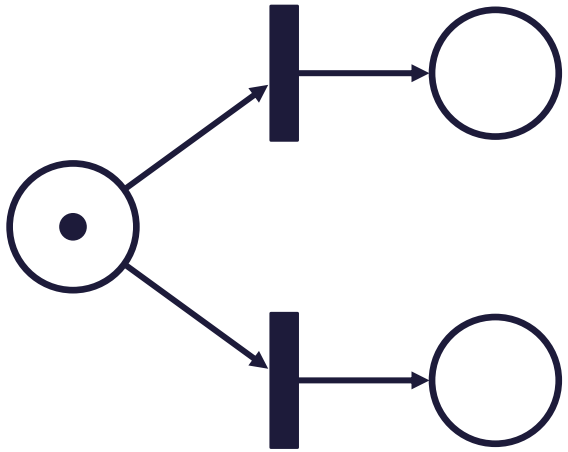
two jobs at once

Join (AND)



wait for both

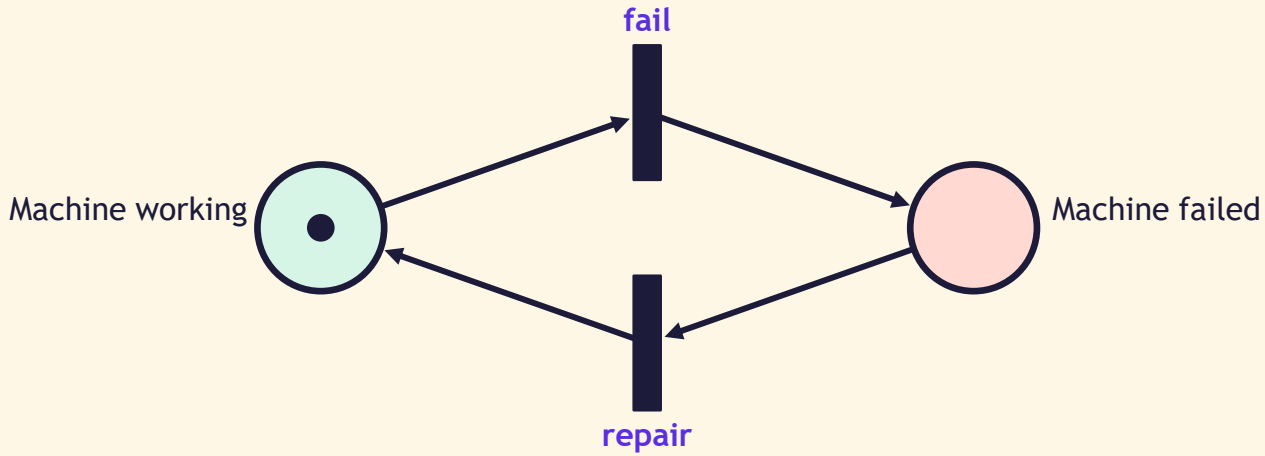
Choice (conflict)



one path only

Example 2: a machine that breaks

⏱ 4 min

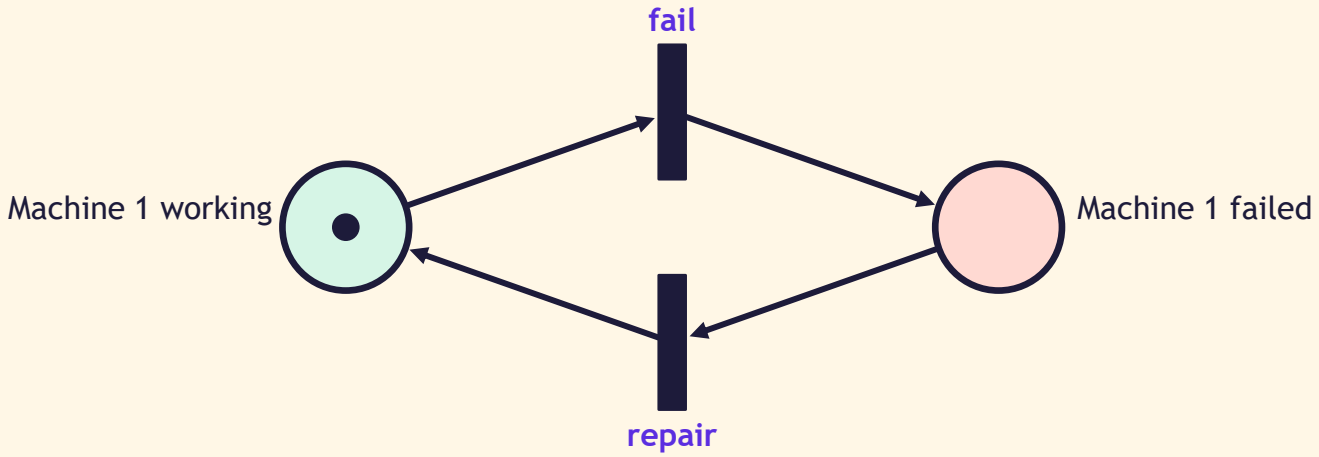


Reading the loop
1 token = the machine is in that state. Exactly one of the two circles holds it, always.

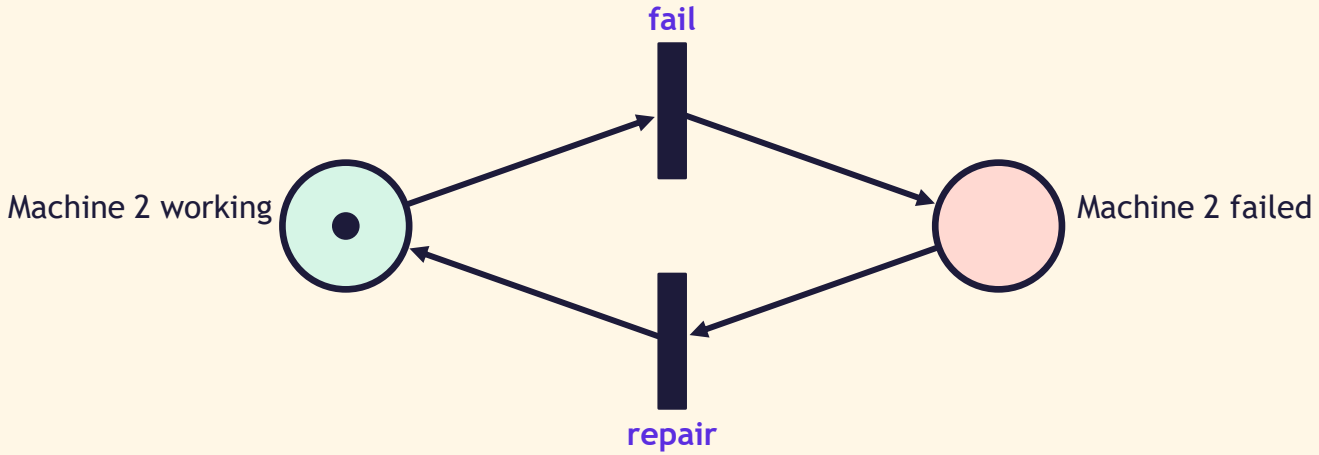
Non-repairable?
Delete the repair bar: the failed state becomes a dead end. Think: a light bulb.

Two machines: hot redundancy

⏱ 4 min



Hot = both active
System is up if at least one "working" circle holds a token.
One failure: zero interruption.

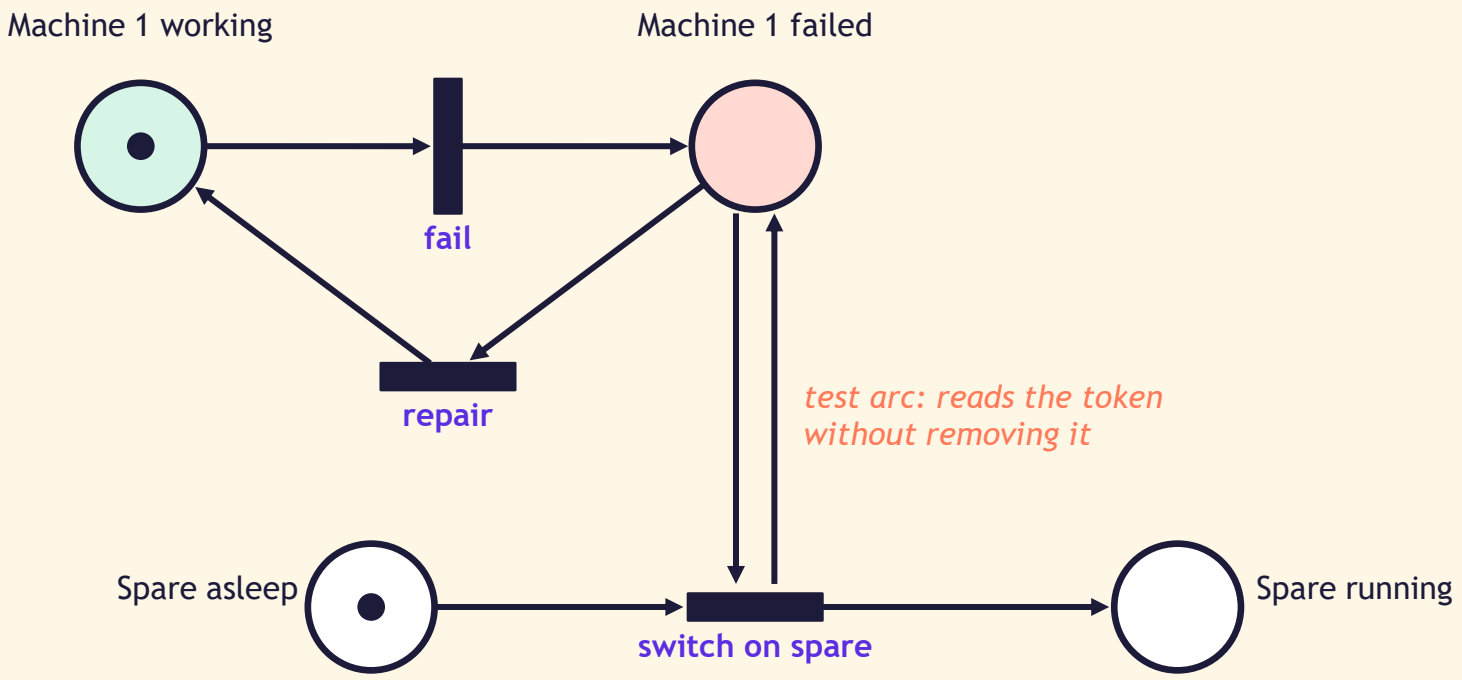


🏭

Two lines, same orders. If one stops, the other keeps shipping.

Two machines: cold redundancy

⏱ 4 min

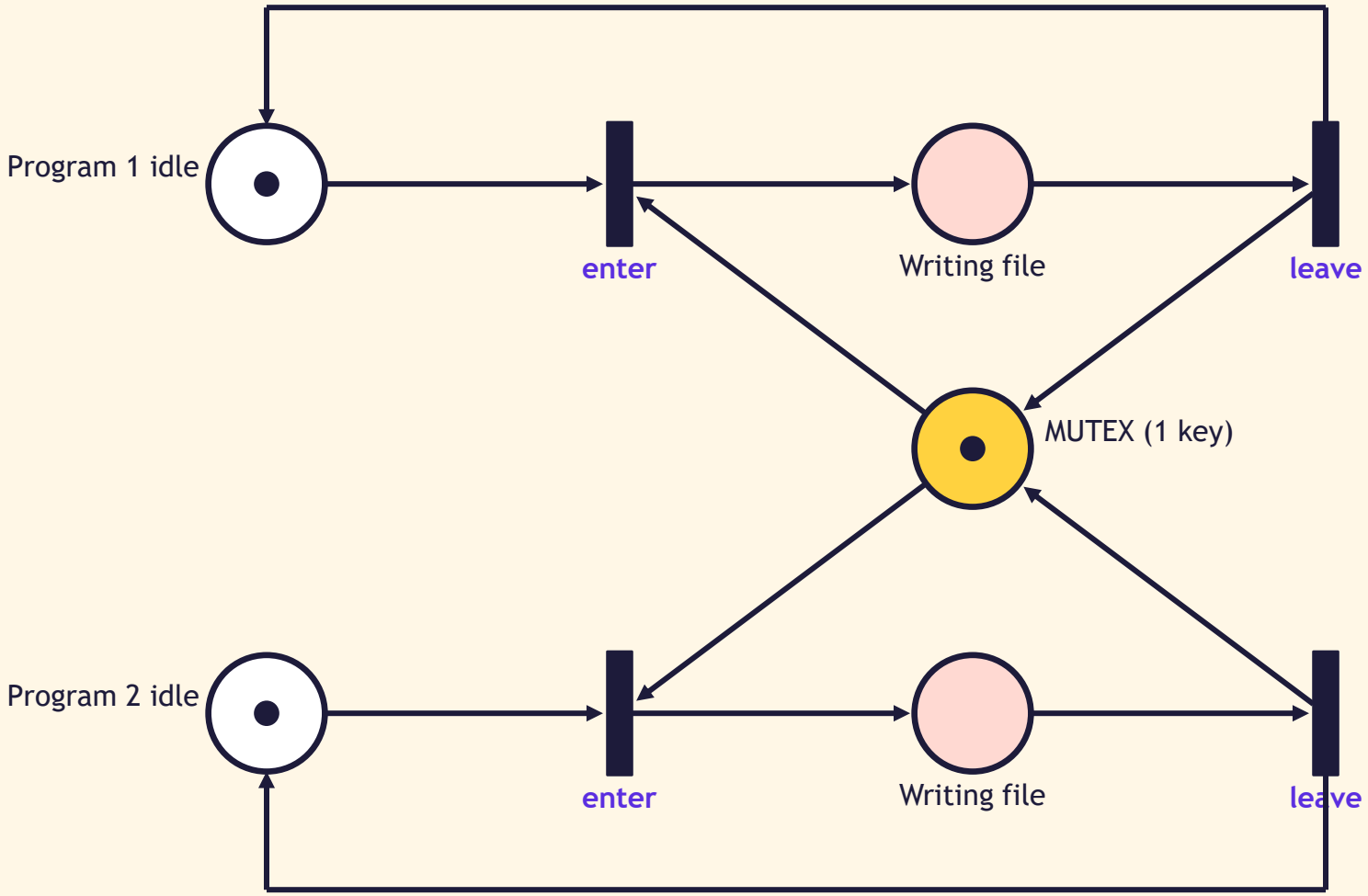


Cold = spare asleep
Spare wakes up only after a failure. Cheaper in energy, but a short startup delay.

Mini challenge
What transition puts the spare back to sleep after repair? (answer in the notes)

Example 3 (IT): two programs, one file

⏱ 5 min

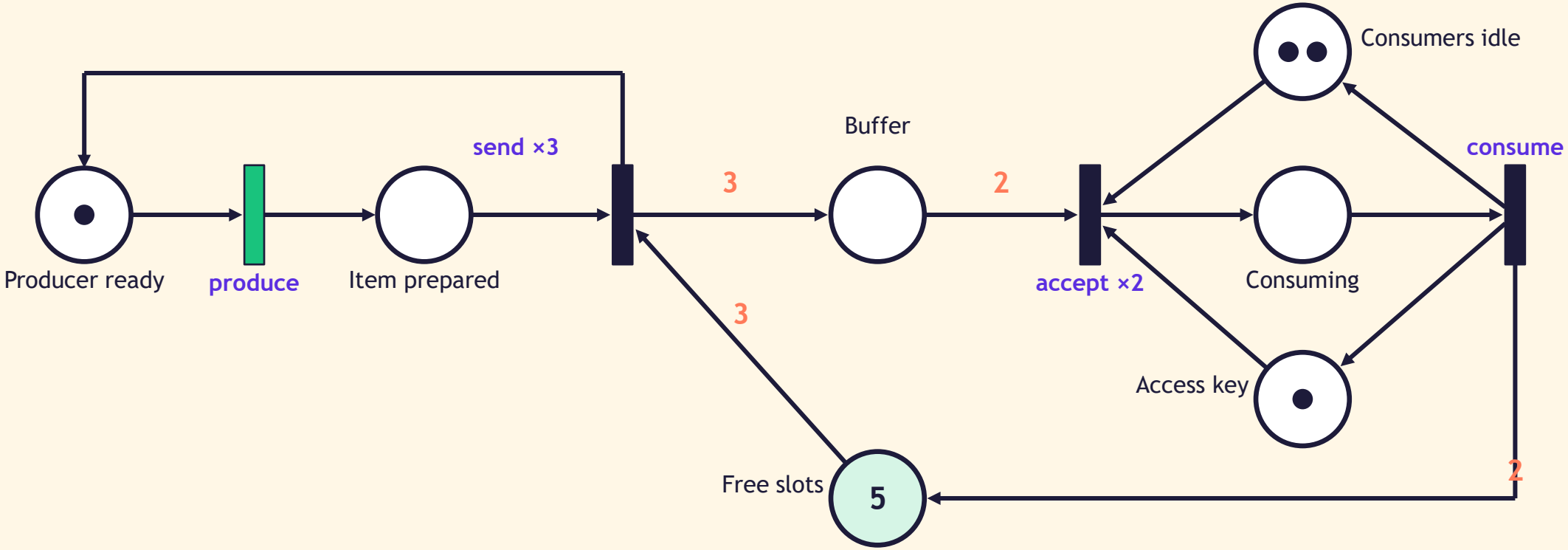


One key only

The mutex token can only be in one place: Program 1 OR Program 2 writes, never both.

Example 4: producer and consumers

⏱ 6 min



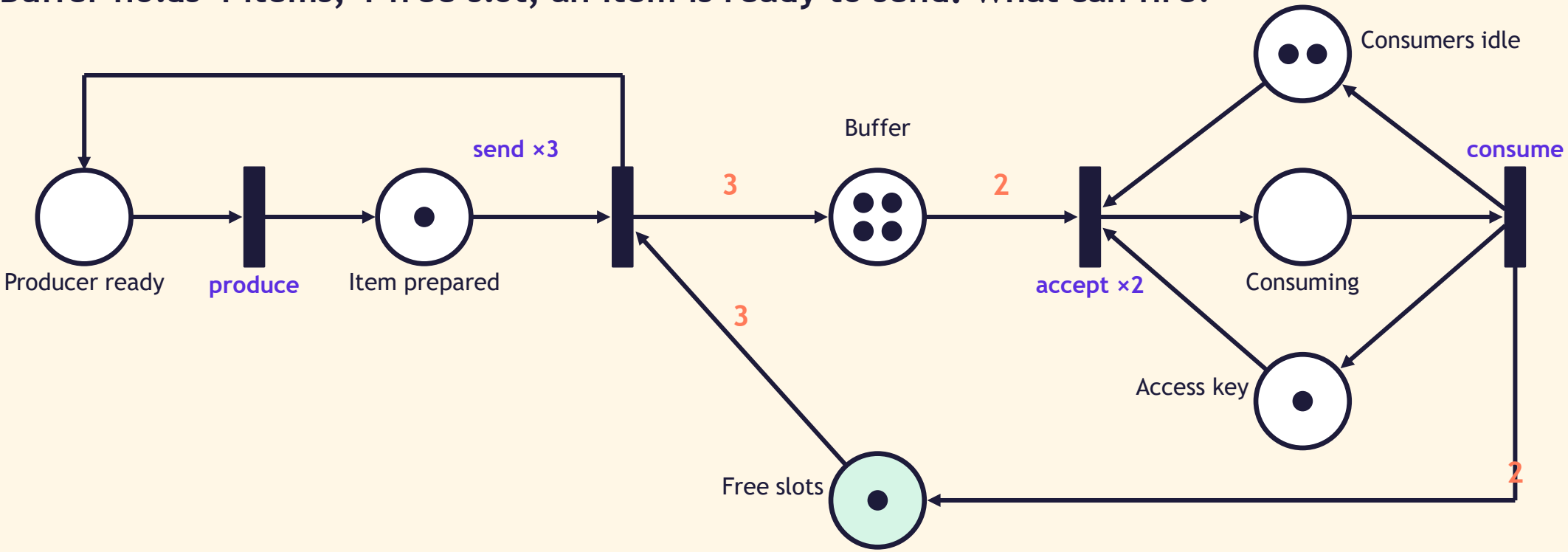
Why the Free slots circle?

Buffer + Free = 5 at all times. Send needs 3 free slots, so it is blocked if only 2 remain. The access key is given back by consume, so the next consumer can enter.

Quiz 2: buffer — your turn

⏱ 4 min

Buffer holds 4 items, 1 free slot, an item is ready to send. What can fire?

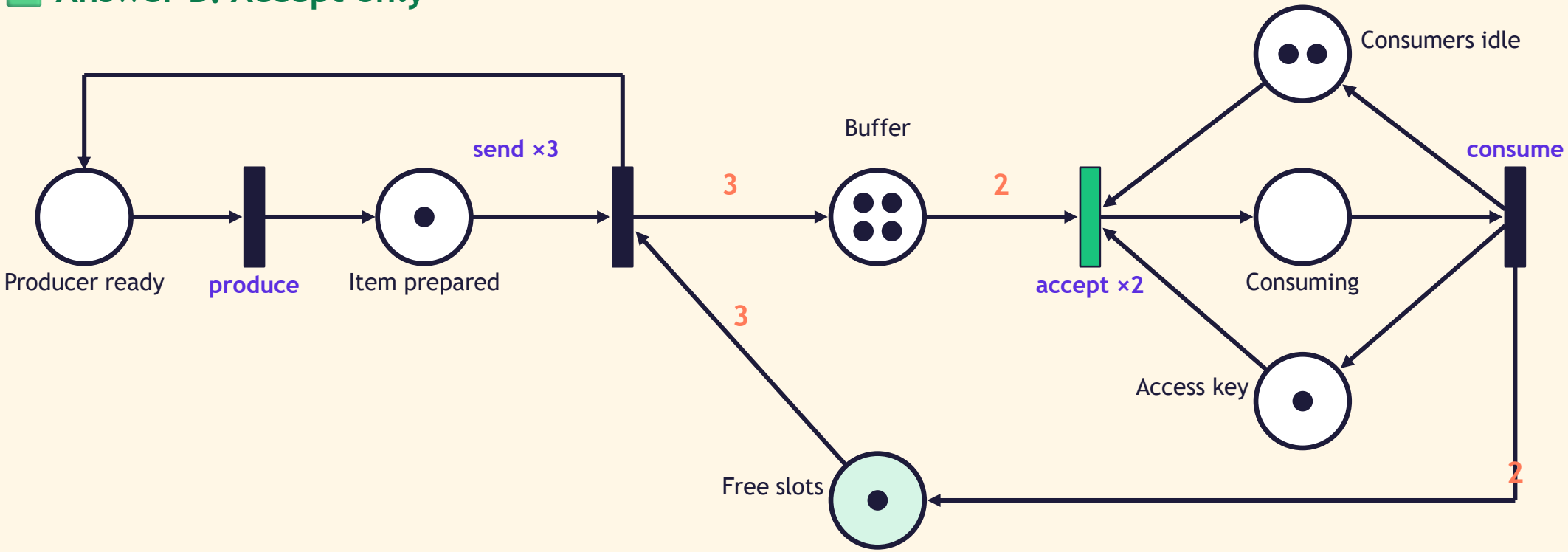


A. Send only | **B. Accept only** | **C. Both** | **D. Nothing**

Quiz 2: buffer — answer

⏱ 0 min

✅ Answer B: Accept only

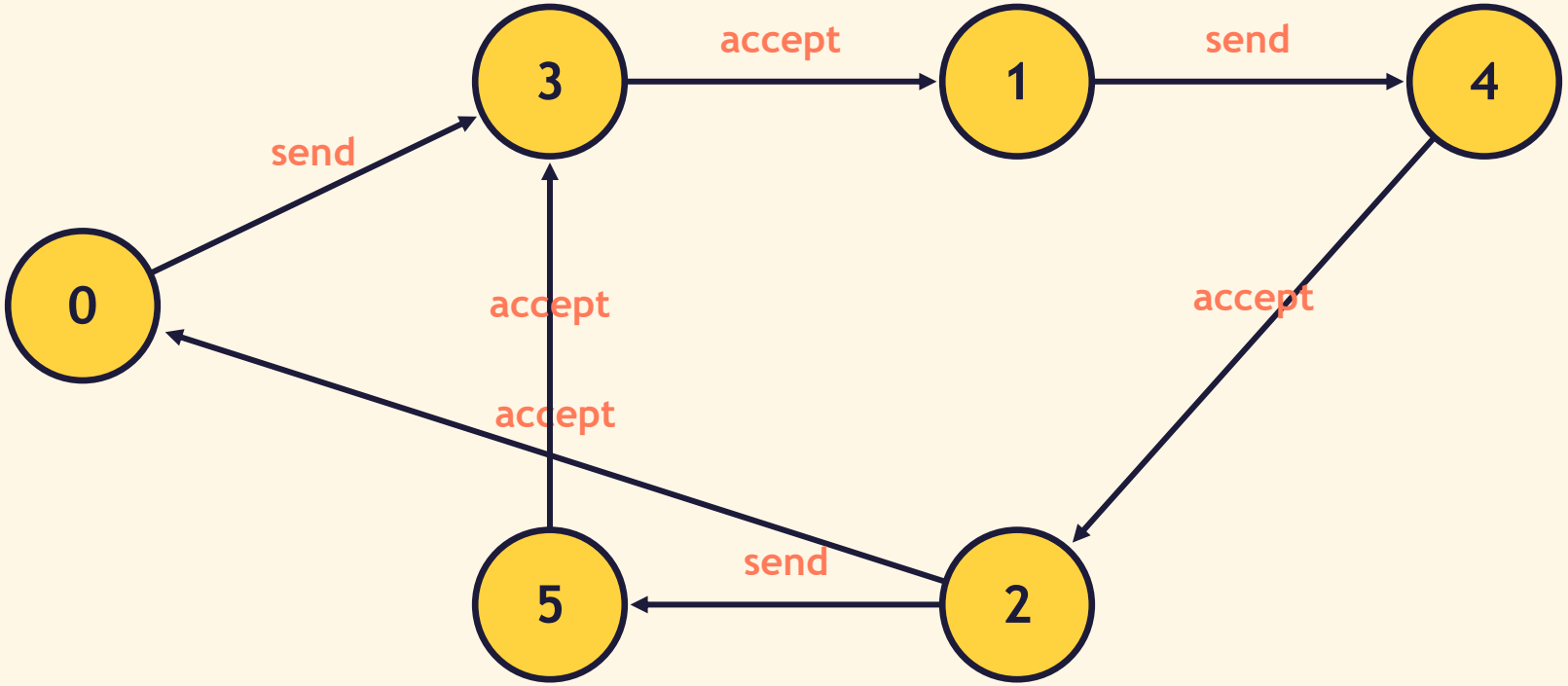


Send needs 3 free slots but only 1 exists: blocked. Accept needs 2 items in the buffer and a free access key: OK. Check: Buffer + Free = 4 + 1 = 5.

All the possible situations

⏱ 4 min

Number = items in the buffer



State map
6 reachable situations.
Every one has an exit: good sign!

Is my design safe? Three questions

⏱ 5 min

Bounded
No place overflows.

Example: the buffer never holds more than 5 items.

Deadlock-free
There is always at least one move.

Example: the clinic always has something to do.

Live (stronger)
Every event can happen again, whatever happened before.

Deadlock-free is not enough: one event could be dead forever.

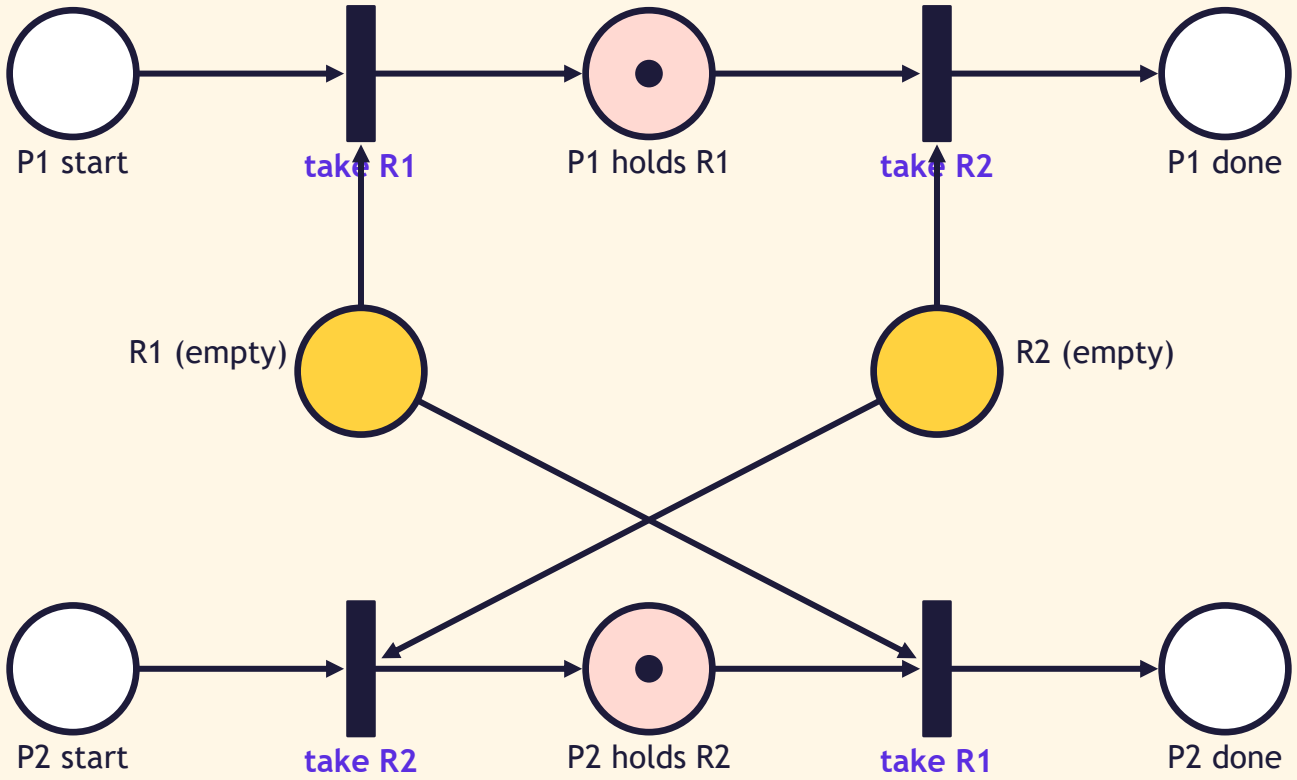
🤔

A hospital with no deadlock but a surgeon who never works again: not live!

Quiz 3: deadlock — your turn

⏱ 4 min

Both programs hold one resource each, R1 and R2 are taken. How many transitions can fire?



The classic trap

P1 takes R1 then R2. P2 takes R2 then R1. Each holds one and waits for the other.

A. One

B. Two

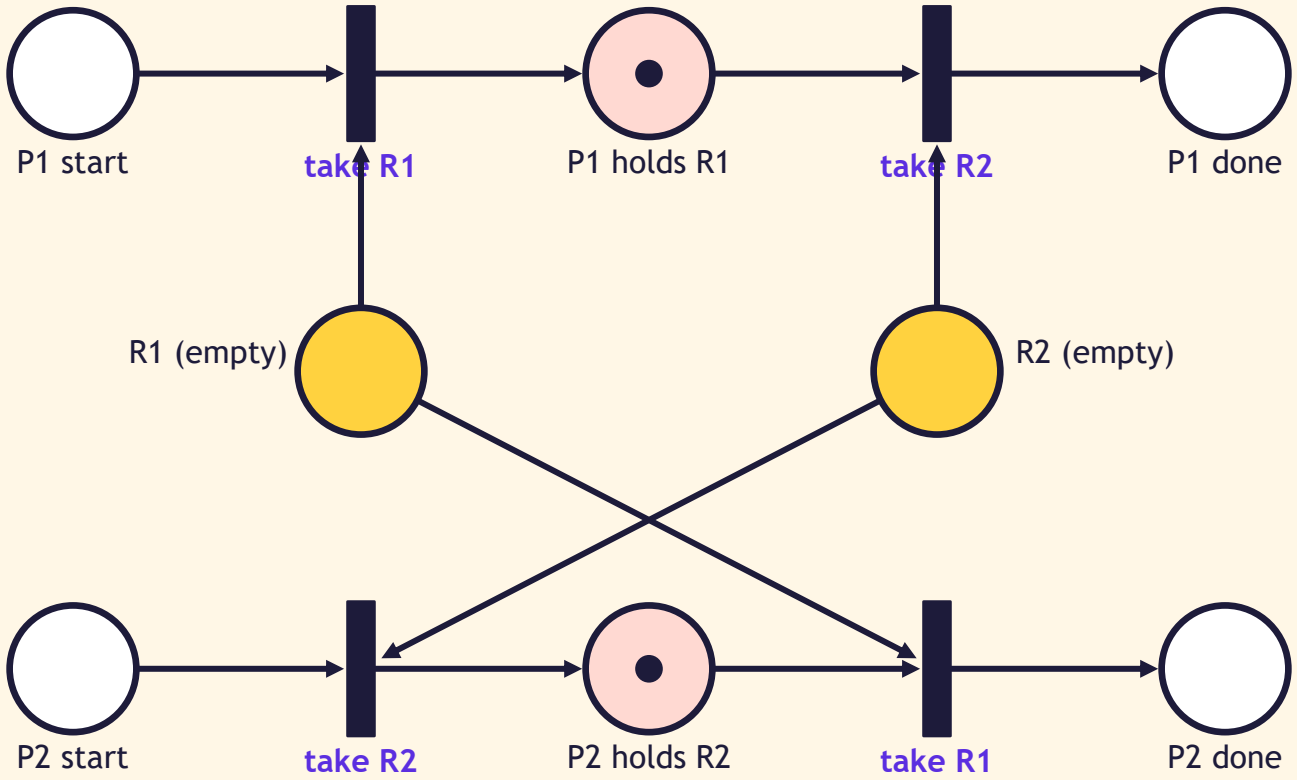
C. All four

D. None: deadlock

Quiz 3: deadlock — answer

⏱ 0 min

✅ **Answer D: None: deadlock**



The classic trap

P1 takes R1 then R2. P2 takes R2 then R1. Each holds one and waits for the other.

Take R2 (P1) needs R2, held by P2. Take R1 (P2) needs R1, held by P1. Nobody can move: a deadlock. Fix: both take R1 first.

Counting tokens: invariants

⏱ 4 min

🏥 Clinic

Doctor free + In consultation = 1

The doctor is never lost or cloned.

💻 Mutex

Writing 1 + Writing 2 + Mutex = 1

At most one program writes: mutual exclusion proved.

🏭 Buffer

Stored + Free = 5

No overflow ever: boundedness proved.

🧮

Count the tokens once. If the total never changes, you have a proof, not a hope.

Adding time and randomness

⏱ 4 min

⏱ Timed
Each transition takes time.
Surgery: 30 minutes.
Assembly: 2 minutes.

🎲 Stochastic
Times are random (rates).
Packet arrivals, machine failures.
This leads straight to queues (next chapter).

🎨 Colored
Tokens carry data: patient age, packet priority, part type.

💻 *Try it: draw your net in ORIS, press simulate, watch the tokens move.*

Your turn: design challenge

⏱ 10 min

🏥 Operating room

2 surgeons, 1 theatre. Surgery needs 1 surgeon and the theatre. Model it.

💻 Shared printer

3 students, 1 printer. Only one can print at a time. Model it, find an invariant.

🏭 Robot arm

1 arm, 2 conveyors. Arm takes a part from either conveyor. Choice or fork?

Checklist (10 min)

1) places and transitions 2) initial marking 3) one invariant 4) can it deadlock?

Recap and exit ticket

⏱ 3 min

Draw
place, bar, arc, token

Fire
enough tokens in, tokens out

Patterns
sequence, fork, join, choice

Check
bounded, deadlock-free, live



Exit ticket: name one thing in your life that is a "shared resource" and draw it as a token.

Questions? Latifa.dekhici@univ-usto.dz